Youngjun Kim, Ph.D.

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EDUCATION

Postdoctoral Scholar, Radiation Oncology, Stanford University, USA **(May. '13 ~ Apr. '14)**

- Deformable surface registration for patient setup in radiotherapy (Advisor: Lei Xing)

Ph.D., Mechanical & Aerospace Engineering, **Seoul National University, Korea** **(Mar. '05 ~ Aug. '09)**

- 3D postoperative analysis of total knee arthroplasty using 2D-3D medical image registration (Advisor: Prof. Kunwoo Lee)

M.S., Mechanical & Aerospace Engineering, Seoul National University, Korea **(Mar. '01 ~ Feb. '03)**

B.S., Mechanical & Aerospace Engineering, Seoul National University, Korea **(Mar. '97 ~ Feb. '01)**

RELEVANT PROFESSIONAL EXPERIENCE

Imagoworks, Inc.

Chief Executive Officer & Founder **(Nov. '19 ~ present)**

Korea Institute of Science and Technology, Seoul, Korea

Principal Researcher at Center for Bionics **(Mar. '18 ~ present)**

Senior Researcher at Center for Bionics **(Sep. '11 ~ Feg. '18)**

Star Postdoctoral Researcher at Intelligence and Interaction Center **(Oct. '09 ~ Aug. '11)**

University of Science and Technology, KIST School, Seoul, Korea

Associate Professor at Division of Bio-Medical Science & Technology **(Mar. '12 ~ present)**

K&I Technology Co., Ltd, Seoul, Korea **(Mar. '03 ~ Aug. '09)**

Chief Researcher at K&I Research Center

Technical environment: C++, MFC, OpenGL, OpenCV, VTK, ITK, CUDA, etc.

PROFESSIONAL MEMBERSHIP

Artificial Intelligence Working Group, **FDI World Dental Federation** **('21 ~ '23)**

Committee Member, **Korean Society for Artificial Intelligence in Medicine** **('18 ~ present)**

Committee Member, **Korean Society of Imaging Informatics in Medicine** **('16 ~ present)**

Committee Member, **Society for Computer Design and Engineering** **(''16 ~ present)**

Committee Member, **Korean Society for Simulation Surgery** **('15 ~ present)**

AWARD

1. Presidential Commendation 2023, Republic of Korea, Dec. 2023
2. Minister's Commendation 2022 for "Contribution to the development of AI industry", Ministry of Science and ICT, Dec. 13, 2022
3. Excellent Technology Award, Society for Computational Design and Engineering, Feb. 2022
4. K-startup Pitching Competition, Ministry of SMEs and Startups (Prize: US $50K), Nov. 20, 2020
5. Winner, Seoul Innovation QuickFire Challenge: Smart Healthcare, awarded by Johnson & Johnson, (two winners among 54 worldwide applicants, Prize: KRW 75M (~US $75K) in grant funding, 1 year residency in Seoul Biohub, mentorship from J&J), Nov. 5, 2019
6. Best Presentation Award, "Web-based deep learning platform for automated 3D modeling of mandible", International Congress of Conference on Computational Design and Engineering 2019, Jul. 9, 2019
7. Minister's Commendation, Ministry of Science and ICT, Dec. 28, 2018
8. Korea Top 100 Excellent Research and Technology Achievement Award in 2018, Ministry of Science and ICT, Oct. 10, 2018
9. Best Presentation Award, "Rotator cuff tear 3D visualization using deep learning", Korea Society of Imaging Informatics in Medicine, Jul. 14, 2018
10. Excellent Researcher Award, Awarded by *National Research Council of Science and Technology*, Feb. 9, 2018
11. KIST Best Researcher of the Month, Sep. 29, 2017
12. Academic Excellence Award, "Serial changes in 3-dimensional supraspinatus muscle volume following rotator cuff repair", The 30th Conference on Korean Orthopaedic Society for Sports Medicine, Sep. 23, 2017
13. Best Poster Award, "Web-based cooperation system for 3D surgical planning", 2017 Summer Conference, Society for Computational Design and Engineering, Aug. 18, 2017
14. Best Paper Award, "Deep reinforcement learning for calibration of 2D-3D registration", Asian Conference on Design and Digital Engineering 2017, Aug. 27, 2017
15. Excellent Research Team Award (Medical IT Team, PI: Laehyun Kim), Awarded by the President of KIST, Feb. 2017
16. Grand Prize, "3D surgical planning software for maxillofacial reconstruction", The 17th Computational Design and Engineering (CDE) Software Contest, Aug. 26, 2016 (Awarded by *Minister of Science, ICT and Future Planning*)
17. Best Paper Award, "Optimal placement of medical robotic system using genetic algorithms", The 1st International Joint Conference on Convergence, Feb. 2015
18. Young Scientist Award, The 2013 Annual Conference of the Society of CAD/CAM Engineers, Jan. 2013
19. Best Paper Award, "Development of cholecystectomy simulation for laparoscopic surgery training", The 2012 Annual Conference of the Society of CAD/CAM Engineers, Feb. 2012
20. Encouragement Prize, The 18th Software Contest of Korea, sponsored by Ministry of Communication and Information, 2006
21. Grand Prize, The 5th CAD/CAM Software Contest of Korea, sponsored by Ministry of Communication and Information, 2002
22. Seoul National University Academic Scholarship, 1998-2000

PUBLICATION

Journal Article

1. Joon Young Lee, Sung Eun Kim, Oh Hyo Kwon, **Youngjun Kim**, Tae-geun Son, Hyuk-Soo Han, Du Hyun Ro, "Shallow trochlear groove and narrow medial trochlear width at the proximal trochlea in patients with trochlear dysplasia: A three-dimensional computed tomography analysis", Vol. 32, No. 6, pp. 1434-1445, (Epub) Apr. 2024
2. Seulgi Lee, Gyujin Choi, Jinhyeok Choi, **Youngjun Kim**, Hee-Kyung Kim, "Effect of high-speed sintering on the marginal and internal fit of CAD/CAM-fabricated monolithic zirconia crowns", *Scientific Reports*, Vol. 13, No. 17215, Oct. 2023
3. Seong-Hun Yoo, Sung-Hwan Choi, Kwang-Mahn Kim, Kee-Joon Lee, **Youngjun Kim**, Jae-Hun Yu, Yeong-Il Choi, Jung-Yul Cha, "Accuracy of 3-dimensional printed bracket transfer tray using an in-office indirect bonding system", *American Journal of Orthodontics & Dentofacial Orthopedics*, Vol. 162, No. 1, pp. 93-102, Jul. 2022
4. Young-Sun Park, Jeong-Ho Choi, **Youngjun Kim**, Sung-Hwan Choi, Ji-Hyun Lee, Kyung-Ho Kim, Chooryung J. Chung, "Deep learning-based prediction of the 3D postorthodontic facial changes", *Journal of Dental Research*, Vol. 101, No. 3, pp. 1372-1379, Jun. 2022
5. Xing-Yu Piao, Ji-Man Park, Hannah Kim, **Youngjun Kim**, Jun-Sung Shim, "Evaluation of difference registration methods and dental restorations on the registration during and accuracy of cone beam computed tomography data and intraoral scans: a retrospective clinical study", *Clinical Oral Investigations*, Vol. 26, pp. 5763-5771, May 2022
6. Seungbin Park, Hannah Kim, Eungjune Shim, Bo-Yeon Hwang, **Youngjun Kim**, Jung-Woo Lee, Hyunseok Seo, "Deep Learning-Based Automatic Segmentation of Mandible and Maxilla in Multi-Center CT Images", *Applied Sciences*, Vol. 12, No. 3, 1358, Jan. 2022
7. Seungbin Park, Hannah Kim, Laehyun Kim, Jin-kuk Kim, In Sik Lee, Ik Hee Ryu*, **Youngjun Kim***, "Artificial Intelligence-Based Nomogram for Small-Incision Lenticule Extraction", *Biomedical Engineering Online*, Vol. 20, No. 38, Apr. 2021
8. Jerome Charton, Stephen Baek, **Youngjun Kim**, "Mesh repairing using topology graphs", *Journal of Computational Design and Engineering*, Vol. 8, No. 1, pp. 251-267, Feb. 2021 (Editor's Choice)
9. Chan-Woo Park, Sung Wook Seo, Noeul Kang, BeomSeok Ko, Byung Wook Choi, Chang Min Park, Dong Kyung Chang, Hwiuoung Kim, Hyunchul Kim, Hyunna Lee, Jinhee Jang, Jong Chul Ye, Jong Hong Jeon, Joon Beom Seo, Kwang Joon Kim, Kyu-Hwan Jung, Namkug Kim, Seungwook Paek, Soo-Yong Shin, Soyoung Yoo, Yoon Sup Choi, **Youngjun Kim**, Hyung-Jin Yoon, "Artificial Intelligence in Health Care: Current Applications and Issues", *Journal of Korean Medical Science*, Vol. 35, No. 42, Nov. 2020
10. Eungjune Shim, Joon Yub Kim, Jong Pil Yoon, Se-Young Ki, Taewoo Lho, **Youngjun Kim***, Seok Won Chung*, "Automated rotator cuff tear classification using 3D convolutional neural network", *Scientific Reports*, Vol. 10, No. 1, pp. 1-9, Sep. 2020
11. Si-Myung Park, Jeonghwan Lee, Seungbin Park, Jung-Woo Lee, Minsoo Park, **Youngjun Kim**, Gunwoo Noh, "Practical bending-angle calculation for an automated surgical plate bending apparatus", *Journal of Mechanical Science and Technology*, Vol. 5, No. 34, pp. 2101-2109, May 2020
12. Hannah Kim, Eungjune Shim, Jungeun Park, Yoon-Ji Kim, Uilyong Lee, **Youngjun Kim**, "Web-based fully automated cephalometric analysis", *Computer Methods and Programs in Biomedicine*, Vol. 194, 105513, Oct. 2020
13. Seungbin Park, Jeonghwan Lee, Si-Myung Park, Gunwoo Noh, Jung-Woo Lee, Min Soo Park, **Youngjun Kim**, "A novel motorized bending apparatus for surgical plates", *Journal of Mechanical Science and Technology*, Vol. 33, No. 8, pp. 3743-3748, Aug. 2019
14. Min ji Kim, Woo Shik Jeong, Hannah Kim, Hyunchul Cho, Yun Hwan Kim, **Youngjun Kim**, Jong Woo Choi, "Clinical application of a novel specific simulation software for three-dimensional orbital volume modeling for orbital wall reconstruction", *Annals of Plastic Surgery*, Vol. 83, No. 1, pp. 48-54,

Jul. 2019

15. Hannah Kim, Tae-geun Son, Jeonghwan Lee, Hyeun A Kim, Hyunchul Cho, Woo Shik Jeong, Jong Woo Choi, **Youngjun Kim**, "Three-dimensional orbital wall modeling using paranasal sinus segmentation", *Journal of Cranio-Maxillo-Facial Surgery*, Vol. 47, No. 6, pp.959-967, Jun. 2019
16. Hannah Kim, Tae-geun Son, Hyunchul Cho, Eungjune Shim, Bo-Yeon Hwang, Jung-Woo Lee, **Youngjun Kim**, "Automated maxillofacial reconstruction software: development and evaluation", *Computer Methods in Biomechanics and Biomedical Engineering: Imaging & Visualization*, doi: 10.1080/21681163.2019.1608308 [Epub ahead of print], May 2019
17. Jeonghwan Lee, Kinde Mekuria, Tea-geun Son, Woo Shik Jeong, Jong Woo Choi, **Youngjun Kim**, "A novel non-invasive patient-specific navigation method for orbital reconstructive surgery: a phantom study using patient data", *Plastic and Reconstructive Surgery*, Vol. 143, No. 3, pp. 602-612, Mar. 2019
18. Seok Won Chung, Seung Seog Han, Ji Whan Lee, Kyung-Soo Oh, Na Ra Kim, Jong Pil Yoon, Joon Yub Kim, Sung Hoon Moon, Jieun Kwon, Hyo-Jin Lee, Young-Min Noh, **Youngjun Kim**, "Automated detection and classification of the proximal humerus fracture by using deep learning algorithm", *Acta Orthopaedica*, Vol. 89, No. 4, pp. 468-473, Aug. 2018
19. Hyung Hwa Jeong, Woo Shik Jeong, Jong Woo Choi, Seung Eun Jeong, Soon Yuhl Nam, Seung Ho Choi, Yoon Sei Lee, Hyeun A Kim, **Youngjun Kim**, "3D computer simulation analysis of the flap volume change in total tongue reconstruction flaps", *Journal of Cranio-Maxillofacial Surgery*, Vol. 46, No. 5, pp. 844-850, May 2018
20. Eungjune Shim, **Youngjun Kim**, Deukhee Lee, Byung Hoon Lee, Sungkyung Woo, Kunwoo Lee, "2D-3D registration for 3D analysis of lower limb", *Applied Mathematics - A Journal of Chinese Universities*, Vol.33, No.1, pp. 59-70, Mar. 2018
21. Ki-Mo Jang, Jong-Hoon Park, Minho Chang, **Youngjun Kim**, Deukhee Lee, Sehyung Park, and Joon Ho Wang, "Three-Dimensional Evaluation of Similarity of Right and Left Knee Joints", *Knee Surgery & Related Research*, Vol.29, No. 4, pp. 307-315, Dec. 2017
22. Yimdon Choi*, **Youngjun Kim***, Eunsoo Park, "Patient-Specific Augmentation Rhinoplasty Using a Three-Dimensional Simulation Program and Three-Dimensional Printing", *Aesthetic Surgery Journal*, **These authors contributed equally to this work*, Vol.37, No. 9, pp. 988-998, Oct. 2017
23. Jerome Charton, Mathieu Laurentjoye, **Youngjun Kim**, "3D Boolean operations in virtual surgical planning", *International Journal of Computer Assisted Radiology and Surgery*, Vol. 12, No. 10, pp. 1697-1709, Oct. 2017
24. Jerome Charton, Laehyun Kim, **Youngjun Kim**, "Boolean operations between two colliding shells: a robust, exact, and simple method", *Journal of Advanced Mechanical Design, Systems, and Manufacturing*, Vol. 11, No. 4, p.JAMDSM0041, Sep. 2017
25. Siyeop Yoon, Sangkyun Shin, Hyunchul Cho, **Youngjun Kim**, Deukhee Lee, Laehyun Kim, Gunwoo Noh, "Enhanced markerless surgical robotic guidance system for keyhole neurosurgery", *Journal of Advanced Mechanical Design, Systems, and Manufacturing*, Vol. 11, No. 4, p.JAMDSM0046, Sep. 2017
26. Seok Won Chung, Kyung-Soo Oh, Sung Gyu Moon, Na Ra Kim, Ji Whan Lee, Eungjune Shim, Sehyung Park, **Youngjun Kim**, "Serial Changes in 3-Dimensional Supraspinatus Muscle Volume following Rotator Cuff Repair", *The American Journal of Sports Medicine*, Vol. 45, No. 10, pp. 2345-2354, Aug. 2017
27. Ho Dong Yu, Jin-Young Park, Jae Hyung Lee, Kyung Soo Oh, Seok Won Chung, Hyunchul Cho, **Youngjun Kim**, "Three-dimensional analysis of scapular kinematics during arm elevation in baseball players with scapular dyskinesis: Comparison of dominant and non-dominant arms", *Asia-Pacific Journal of Sports Medicine, Arthroscopy, Rehabilitation and Technology*, Vol. 9, pp. 115-116, Jul. 2017

28. Gayoung Kim, Eungjune Shim, **Youngjun Kim***, Yong Oock Kim*, "Virtual Reality and 3D Printing for Craniopagus Surgery", *Journal of International Society for Simulation Surgery*, Vol. 4, No. 1, pp. 9-12, Jun. 2017
29. **Youngjun Kim**, Hannah Kim, Yong Oock Kim, "Virtual Reality and Augmented Reality in Plastic Surgery: A Review", *Archives of Plastic Surgery*, Vol. 44, No. 3, pp. 179-187, May 2017
30. Sunhee Kim, Deukhee Lee, Sehyung Park, Kyung-Soo Oh, Seok Won Chung, **Youngjun Kim**, "Automatic segmentation of supraspinatus from MRI by internal shape fitting and autocorrection", *Computer Methods and Programs in Biomedicine*, Vol. 140, pp. 165-174, Mar. 2017
31. **Youngjun Kim**, Byung Hoon Lee, Kinde Mekuria, Hyungchul Cho, Sehyung Park, Joon Ho Wang, Deukhee Lee, "Registration accuracy enhancement of a surgical navigation system for anterior cruciate ligament reconstruction: A phantom and cadaveric study", *The Knee*, Vol.24, No. 2, pp. 329-339, Feb. 2017
32. Chang Kyu Lee, **Youngjun Kim**, Nam Lee, Byungwoo Kim, Do Young Kim, Seong Yi, "Feasibility study of utilization of action camera, GoPro Hero 4, Google glass and Panasonic HX-A100 in Spine surgery", *Spine*, Vol. 42, No. 4, pp. 275-280, Feb. 2017
33. Quoc Cuong Nguyen, **Youngjun Kim**, HyukDong Kwon, "Optimization of layout and path planning of surgical robotic system", *International Journal of Control, Automation and Systems*, Vol. 15, No. 1, pp. 375-384, Jan. 2017
34. Quoc Cuong Nguyen*, **Youngjun Kim***, Sehyung Park, HyukDong Kwon, "End-effector path planning and collision avoidance for robot-assisted surgical system", *International Journal of Precision Engineering and Manufacturing*, **These authors contributed equally to this work*, Vol. 17, No. 12, pp. 1703-1709, Dec. 2016
35. Byung Hoon Lee, Dong Ho Kum, Im Joo Rhyu, **Youngjun Kim**, Hyunchul Cho, Joon Ho Wang, "Clinical advantages of image‐free navigation system using surface‐based registration in anatomical anterior cruciate ligament reconstruction", *Knee Surgery, Sports Traumatology, Arthroscopy*, Vol. 24, No. 11, pp. 3556-3564, Nov. 2016
36. Jai Gon Seo*, Sang Min Kim, Jung Min Shin, **Youngjun Kim***, Byung Hoon Lee, "Safety of simultaneous bilateral total knee arthroplasty using an extramedullary referencing system: results from 2098 consecutive patients", *Archives of Orthopaedic and Trauma Surgery*, **These authors contributed equally to this work*, Vol. 136, No. 11, pp. 1615-1621, Nov. 2016
37. **Youngjun Kim**, Yong Hum Na, Lei Xing, Rena Lee, Sehyung Park, "Automatic deformable surface registration for medical applications by radial basis function-based robust point-matching", *Computers in Biology and Medicine*, Vol. 77, No. 1, pp. 173-181, Oct. 2016
38. Sin Hyung Park, Sang Won Moon, Byung Hoon Lee, Sehyung Park, **Youngjun Kim**, Deukhee Lee, Sunghwan Lim, Joon Ho Wang, "Arthroscopically blind anatomical anterior cruciate ligament reconstruction using only navigation guidance: a cadaveric study", *The Knee*, Vol. 23, No. 5, Oct. 2016
39. **Youngjun Kim**, Wontae Kim, Deukhee Lee, "3D Inspection by Registration of CT and Dual X-ray Images", *Journal of International Society for Simulation Surgery*, Vol. 3, No. 1, pp. 16-21, Jun. 2016
40. Jong Pil Yoon, Seok Won Chung, Ju-Eun Kim, Hyung Sup Kim, Hyun-Joo Lee, Won-Ju Jeong, Kyung-Soo Oh, Dong-Oh Lee, Anna Seo, **Youngjun Kim**, "Intra-articular injection, subacromial injection, and hydrodilatation for primary frozen shoulder: a randomized clinical trial", *Journal of Shoulder and Elbow Surgery*, Vol. 25, No. 3, pp. 376-383, Mar. 2016
41. Kinde Mekuria, **Youngjun Kim**, Hyunchul Cho, Deukhee Lee, Sehyung Park, Byung Hoon Lee, Ki-Mo Jang, Joon Ho Wang, "The Effect of Optical Marker Configuration on Tracking Accuracy in Image Guided Surgery", *Studies in Health Technology and Informatics*, Vol. 220, pp. 227-232, Feb. 2016

42. **Youngjun Kim**, Laehyun Kim, Deukhee Lee, Sangkyun Shin, Hyunchul Cho, Frederick Roy, Sehyung Park, "Deformable mesh simulation for virtual laparoscopic cholecystectomy training", *Visual Computer*, Vol. 31, No. 4, pp. 485-495, Apr. 2015
43. **Youngjun Kim**, Ruijiang Li , Yong Hum Na , Lei Xing, Rena Lee, "Accuracy of surface registration compared to conventional volumetric registration in patient positioning for head-and-neck radiotherapy: a simulation study using patient data", *Medical Physics*, Vol. 41, No. 12, Dec. 2014
44. Sangkyun Shin, **Youngjun Kim**, Hyunchul Cho, Deukhee Lee, Sehyung Park, Laehyun Kim, Gerard Kim, "A single camera tracking system for 3D position, grasper angle, and rolling angle of laparoscopic instruments", *International Journal of Precision Engineering and Manufacturing*, Vol. 15, No. 10, pp. 2155-2160, Oct. 2014
45. Jin hyeok Choi, Seung-Yeob Baek, Tae-geun Son, **Youngjun Kim**, Kunwoo Lee, "Automatic Detection of Inferior Alveolar Nerve Canal from Cone Beam Computed Tomography for Dental Surgery Planning", *Studies in Health Technology and Informatics*, Vol. 196, pp. 61-65, Feb., 2014
46. **Youngjun Kim,** DongJune Chang, Jungsik Kim, Sehyung Park, "Gallbladder removal simulation for laparoscopic surgery training: a hybrid modeling method", *Journal of Computer Science Technology*, Vol. 28, No. 3, pp. 499-507, May 2013
47. Jin hyeok Choi, **Youngjun Kim**, Tae-kyoung Yi, Jekyo Jung, Yong Kim, Sehyung Park, "Optimized Marker for Template-Guided Intraoral Surgery", *Studies in Health Technology and Informatics*, Vol. 184, pp. 85-91, 2013
48. **Youngjun Kim**, Seungbin Lee, Frederick Roy, Deukhee Lee, Laehyun Kim, Sehyung Park, "Carving Mesh with Deformation for Soft Tissue Removal Simulation", *Mesh Processing in Medical Image Analysis, Lecture Notes in Computer Science*, Vol. 7599, pp. 70-79, Oct. 2012
49. **Youngjun Kim,** Wontae Kim, Sehyung Park, Jong-Hyeong Kim, "Calibration method for micro-scale dual X-ray imaging system", *International Journal of Precision Engineering and Manufacturing*, Vol. 13, No. 6, pp. 877-882, Jun. 2012
50. **Youngjun Kim**, Kang-Il Kim, Jin Hyeok Choi, Kunwoo Lee, "Novel methods for 3D postoperative analysis of total knee arthroplasty using 2D–3D image registration", *Clinical Biomechanics*, Vol. 26, No. 4, pp. 384-391, May 2011
51. Sangkyun Shin, **Youngjun Kim**, Hyunsoo Kwak, Deukhee Lee, Sehyung Park, "3-D tracking of surgical instruments using a single camera for laparoscopic surgery simulation", *Studies in Health Technology and Informatics*, Vol. 163, pp. 581-587, Feb. 2011
52. Daekwang Myung, **Youngjun Kim**, Jin hyeok Choi, Kunwoo Lee, "Scaled attenuation fields: improved real-time generation method for digitally reconstructed radiographs", *International Journal of Precision Engineering and Manufacturing*, Vol. 11, No. 5, pp. 791-798, Oct. 2010
53. **Youngjun Kim**, Kunwoo Lee, Wontae Kim, "3D virtual simulator for breast plastic surgery", *Computer Animation and Virtual Worlds*, Vol. 19, Issue 3-4, pp. 515-526, Aug. 2008
54. **Youngjun Kim**, Jungbum Cho, Bohyun Kim, Kunwoo Lee, "Modeling of human head for custom wig production", *Lecture Notes in Computer Science*, Vol. 4561, pp. 874-883, Springer Berlin, Jul. 2007

Book

1. **Youngjun Kim**, Hyunchul Cho, Jin hyeok Choi, "VTK 프로그래밍(VTK programming)", 일진사 (Iljinsa co.), ISBN 978-89-429-1525-5, Sep. 2017

White Paper

1. Falk Schwendicke, Markus Blatz, Sergio Uribe, William Cheung, Mahesh Verma, Jina Linton, and **Youngjun Kim**, "Artificial Intelligence for Dentistry", FDI Artificial Intelligence Working Group, Feb.

2023

2. "Artificial Intelligence in Health Care: Current Applications and Issues", Korean Society of Artificial Intelligence in Medicine, ISBN 979-11-6003-205-5, Feb. 2020

International Conference Paper

1. Eunsoo Jeon, Eungjune Shim, Jangwoo An, Jungwoo Lee, **Youngjun Kim**, "Web-based deep learning platform for automated 3D modeling of mandible", International Congress of Conference on Computational Design and Engineering 2019, Jul. 9, 2019
2. Taeseok Lee, Yong-Soo Kim, Tae-Geun Son, **Youngjun Kim**, "Development of surgical assistant navigation system using Augmented Reality glasses", *Asian Conference on Design and Digital Engineering*, Okinawa, Japan, Nov., 2018
3. Yong-Soo Kim, EungJune Shim, Taeseok Lee, **Youngjun Kim**, "Development of an augmented reality application for 3D object manipulation", *Asian Conference on Design and Digital Engineering*, Okinawa, Japan, Nov., 2018
4. Seungbin Park, Si-Myung Park, Jeonghwan Lee, Tae-geun Son, Jung-Woo Lee, Minsoo Park, Gunwoo Noh, **Youngjun Kim**, "Automatic bending device for surgical plate", *International Symposium on Precision Engineering and Sustainable Manufacturing (PRESM2018)*, Sapporo, Japan, Jul. 2018
5. Tae-geun Son, Yong-soo Kim, Hyunchul Cho, **Youngjun Kim**, "Development of 3D printing simulation system for web-based 3D printing service platform", *International Symposium on Precision Engineering and Sustainable Manufacturing (PRESM2018)*, Sapporo, Japan, Jul. 2018
6. Seungbin Park, Jeonghwan Lee, Si-Myung Park, Gunwoo Noh, Jung-Woo Lee, Min Soo Park, **Youngjun Kim**, "A Novel Motorized Bending Apparatus for Surgical Plate", *Emerging Technologies in Mechanical Engineering*, Jeju, Korea, Jul., 2018
7. Tae-geun Son, Hannah Kim, Hyunchul Cho, Jung-Woo Lee, **Youngjun Kim**, Sehyung Park, "Evaluation of surgical planning software for mandibular reconstruction using fibular free flap", *International Journal of Computer Assisted Radiology and Surgery 2018*, Barcelona, Spain, Jun., 2018
8. **Youngjun Kim**, "3D printing for medical application", Tutorial Session, *International Symposium on Computational Design & Engineering 2017*, Ho Chi Minh, Vietnam, Dec. 2017
9. Tae-geun Son, **Youngjun Kim**, "Reassembly of broken objects using surface signatures and its evaluation", *International Symposium on Computational Design & Engineering 2017*, Ho Chi Minh, Vietnam, Dec. 2017
10. EungJune Shim, Hannah Kim, Laehyun Kim, **Youngjun Kim**, "Deep reinforcement learning for calibration of 2D-3D registration", *Asian Conference on Design and Digital Engineering (ACDDE) 2017*, Changsha, China, Aug., 2017
11. **Youngjun Kim**, Hannah Kim, Hyunchul Cho, Jerome Charton, Jung-Woo Lee, Laehyun Kim, "Development of surgical planning software for mandibular reconstruction using fibular flap", *International Journal of Computer Assisted radiology and Surgery 2017*, Barcelona, Spain, Jul., 2017
12. Hannah Kim, **Youngjun Kim**, Hyunchul Cho, Woo Shik Jeong, Jong Woo Choi, Sehyung Park, "Three-dimensional modeling, preoperative diagnosis, and patient specific implant design for orbital wall reconstruction", *International Journal of Computer Assisted Radiology and Surgery 2017*, Barcelona, Spain, Jul., 2017
13. Siyeop Yoon, Jinwon Son, **Youngjun Kim**, Sehyung Park, Deukhee Lee, "A study for non-rigid 2x2D-3D registration of coronary artery images using bifurcation points matching with bi-plane x-ray fluoroscopy", *The 14th International Conference on Ubiquitous Robots and Ambient Intelligence (URAI)*, Jeju, Korea, Jun., 2017

14. **Youngjun Kim**, Hyeun A Kim, Hyunchul Cho, Sunhee Kim, Jerome Charton, Deukhee Lee, Sehyung Park, Woo Shik Jeong, Jong Woo Choi, "3D surgical planning and analysis software for orbital fracture reconstruction", *International Forum on Medical Imaging in Asia (IFMIA) 2017*, Okinawa, Japan, Jan., 2017
15. **Youngjun Kim**, Hyunchul Cho, Hannah Kim, Hyeun A Kim, Jerome Charton, Deukhee Lee, Laehyun Kim, Sehyung Park, Jung-Woo Lee, Jong Woo Choi, "Virtual simulation software for craniomaxillofacial reconstruction surgery", *The 9th Congress of International Society*, Nara, Japan, Dec., 2016
16. EungJun Shim, Sehyung Park, **Youngjun Kim**, Byung Hun Lee, "3-D analysis method of rotation angle difference between ideal and actual anterior-posterior x-ray images of femur in standing posture", *2016 BMES Annual Meeting*, Minneapolis, USA, Oct., 2016
17. Sungkyung Woo, **Youngjun Kim**, Jeonghwan Lee, Kunwoo Lee, "Efficient registration methods between 2D X-ray and 3D CT data of different part of human skeleton", *Asian Conference on Design and Digital Engineering*, Jeju, Korea, Oct. 2016
18. **Youngjun Kim**, Sunhee Kim, Hannah Kim, Hyunchul Cho, Deukhee Lee, Laehyun Kim, Sehyung Park, Jung-Woo Lee, "Automatic segmentation and user-friendly software techniques for virtual surgical planning of mandibular reconstruction", *IEEE Engineering in Medicine and Biology Society*, Orlando, USA, Aug., 2016
19. Si-Myung Park, Deukhee Lee, **Youngjun Kim**, Jung-Woo Lee, Gunwoo Noh, "Stability of the permanently bent plates used in mandibular reconstruction surgery", *IEEE Engineering in Medicine and Biology Society*, Orlando, USA, Aug., 2016
20. **Youngjun Kim**, Sunhee Kim, Hannah Kim, Hyunchul Cho, Deukhee Lee, Laehyun Kim, Sehyung Park, Jung-Woo Lee, "Virtual Surgery Planning and Surgical Guide Design using Automatic Active Contour Segmentation for Maxillofacial Reconstruction Surgery", *International BioMedical Engineering Conference*, Kyungju, Korea, Nov. 2015
21. Sunhee Kim, **Youngjun Kim**, Deukhee Lee, and Sehyung Park, "Active Contour Segmentation Using Level Set Function with Enhanced Image from Prior Intensity", *IEEE Engineering in Medicine and Biology Society*, Milan, Italy, Aug., 2015
22. Sangkyun Shin, Hyunchul Cho, Siyeop Yoon, Kyusic Park, **Youngjun Kim**, Sehyung Park, Laehyun Kim, Deukhee Lee, "Markerless Surgical Robotic System for Intracerebral Hemorrhage Surgery", *IEEE Engineering in Medicine and Biology Society*, Milan, Italy, Aug., 2015
23. Anna Seo, Sunhee Kim, **Youngjun Kim**, Laehyun Kim, Sehyung Park, Suk Won Jung, "Three-Dimensional Visualization for Surgical Planning of Rotator Cuff Tears", *The Asian Conference on Computer Aided Surgery 2015*, Singapore, Jul., 2015
24. **Youngjun Kim**, Hyunchul Cho, Kinde Mekuria, Deukhee Lee, Sehyung Park, Byung Hoon Lee, Dong Ho Kum, Joon Ho Wang, "Development of a surgical navigation system for Development of a surgical navigation system for anterior cruciate ligament reconstruction", *International Journal of Computer Assisted radiology and Surgery 2015*, Barcelona, Spain, Jun., 2015
25. Quoc Cuong Nguyen, **Youngjun Kim**, HyukDong Kwon, "Optimal placement of medical robotic system using genetic algorithms", *The 1st International Joint Conference on Convergence*, Ho Chi Minh, Vietnam, Feb., 2015
26. Kinde Mekuria, **Youngjun Kim**, Hyunchul Cho, Deukhee Lee, Sehyung Park, Eunsu Lee, Byung Hun Lee, Joon Ho Wang, "Surgical navigation system for anterior cruciate ligament reconstruction", *International Biomedical Engineering Conference 2014*, Gwangju, Korea, Nov., 2014
27. Quoc Cuong Nguyen, **Youngjun Kim**, Sehyung Park, HyukDong Kwon, "Layout optimization of medical robotic system in the operating room", *International Biomedical Engineering Conference 2014*, Gwangju, Korea, Nov., 2014

28. **Youngjun Kim**, Ruijiang Li, Yong Hum Na, Cesare Jenkins, Rena Lee, Lei Xing, "Is surface registration accurate enough for patient setup in head-and-neck radiation therapy?", *American Society of Radiation Therapy's 56th Annual Meeting*, San Francisco, USA, Sep., 2014
29. Yong Hum Na, Daniel Kapp, **Youngjun Kim**, Tae-Suk Suh, Lei Xing, "VMAT Treatment Planning Using Cloud Computing", *American Society of Radiation Therapy's 56th Annual Meeting*, San Francisco, USA, Sep., 2014
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